LAPORAN PRAKTIKUM 1

SISTEM OPERASI

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2025

BAB 1

PENDAHULUAN

1.1 LATAR BELAKANG

Sistem operasi merupakan komponen utama dalam setiap perangkat komputer yang berfungsi sebagai penghubung antara pengguna dengan perangkat keras. Tanpa sistem operasi, pengguna tidak dapat menjalankan aplikasi atau mengakses sumber daya komputer secara efisien. Linux sebagai salah satu sistem operasi open-source banyak digunakan dalam dunia akademik, industri, maupun penelitian karena kestabilan, keamanan, dan fleksibilitasnya. Melalui praktikum Sistem Operasi berbasis Linux, mahasiswa dapat memahami secara langsung konsep dasar sistem operasi seperti manajemen proses, manajemen memori, sistem berkas, dan mekanisme komunikasi antara aplikasi dengan kernel melalui system calls. Selain itu, praktikum ini juga memberikan pengalaman dalam memahami struktur internal sistem operasi, mulai dari layanan sistem (system services), pembuatan kernel module, hingga proses booting dan debugging. Pemahaman terhadap konsep dan praktik ini sangat penting karena sistem operasi merupakan fondasi utama bagi pengembangan perangkat lunak dan sistem komputer yang andal. Dengan mempelajari Linux secara praktis, mahasiswa diharapkan mampu menganalisis, memodifikasi, dan mengelola sistem operasi secara efisien, serta memiliki keterampilan teknis yang dapat diterapkan dalam lingkungan kerja profesional di bidang teknologi informasi.

1.2 RUMUSAN MASALAH

1. Bagaimana sistem operasi Linux mengelola proses, memori, dan sistem berkas secara efisien?
2. Bagaimana system calls dan system services bekerja sebagai penghubung antara aplikasi dan kernel?
3. Bagaimana proses debugging dan analisis performa dilakukan untuk mendeteksi serta memperbaiki kesalahan pada sistem operasi?

1.3 TUJUAN

1. Memahami konsep dasar dan fungsi utama sistem operasi Linux dalam pengelolaan sumber daya komputer.
2. Mempelajari cara kerja system calls, system services, serta struktur internal Linux melalui praktik langsung.
3. Menguasai teknik debugging dan analisis sistem guna meningkatkan pemahaman terhadap kinerja serta stabilitas sistem operasi.

BAB II

PEMBAHASAN

2.1 Operating-System Services

Operating-System Services (layanan sistem operasi) adalah fungsi utama yang disediakan oleh sistem operasi untuk mendukung eksekusi program dan pengelolaan sumber daya komputer. Dalam praktik Linux ini, perintah seperti `uname -a`, `ps aux`, dan `free -h` digunakan untuk melihat informasi sistem, proses, memori, CPU, serta perangkat penyimpanan. Program C `os_services.c` menunjukkan berbagai layanan OS seperti program execution (menjalankan proses dan menampilkan PID/PPID), I/O operations (membaca/menulis file), file system manipulation (memeriksa atribut file), communications (menampilkan hostname), error detection (menangani kesalahan file), dan resource allocation (mengatur penggunaan memori). Semua layanan ini bekerja untuk memastikan sistem berjalan stabil, efisien, dan aman.

2.1.1 Menjelajahi Layanan OS

- **Lihat Informasi Sistem**

```
danyp@LAPTOP-GRJL2V1D:~$ uname -a
Linux LAPTOP-GRJL2V1D 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/L
inux
danyp@LAPTOP-GRJL2V1D:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
```

- **Lihat proses yang berjalan**

```
danyp@LAPTOP-GRJL2V1D:~$ ps aux | head -20
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  21772 12648 ?        Ss   20:46   0:01 /sbin/init
root           2  0.0  0.0   3072  1664 ?        Sl   20:46   0:00 /init
root           6  0.0  0.0   3072  1792 ?        Sl   20:46   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
--log-truncate
root          43  0.0  0.3  50432 15488 ?        S<s  20:46   0:00 /usr/lib/systemd/systemd-journald
root          97  0.0  0.1  25536  6656 ?        Ss   20:46   0:00 /usr/lib/systemd/systemd-udevd
systemd+     108  0.0  0.3  21456 12672 ?        Ss   20:46   0:00 /usr/lib/systemd/systemd-resolved
systemd+     112  0.0  0.2  91024  7808 ?        Ssl  20:46   0:00 /usr/lib/systemd/systemd-timesyncd
root         161  0.0  0.0   4236  2560 ?        Ss   20:46   0:00 /usr/sbin/cron -f -P
message+     162  0.0  0.1   9528  4992 ?        Ss   20:46   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
stemd-activation --syslog-only
root         169  0.0  0.2  17968  8576 ?        Ss   20:46   0:00 /usr/lib/systemd/systemd-logind
root         182  0.0  0.3 1756096 12672 ?       Ssl  20:46   0:00 /usr/libexec/wsl-pro-service -vv
syslog       185  0.0  0.1 222508  5504 ?        Ssl  20:46   0:00 /usr/sbin/rsyslogd -n -iNONE
root         189  0.0  0.0   3160  1792 hvc0     Ss+  20:46   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600
 vt220
root         193  0.0  0.0   3116  1920 tty1     Ss+  20:46   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root         198  0.0  0.5 107012 22528 ?        Ssl  20:46   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
-shutdown --wait-for-signal
root         285  0.0  0.0   3076   896 ?        Ss   20:46   0:00 /init
root         286  0.0  0.0   3092  1152 ?        S    20:46   0:00 /init
danyp        287  0.0  0.1   6072  5248 pts/0    Ss   20:46   0:00 -bash
root         288  0.0  0.0      0     0 ?        Zs   20:46   0:00 [SessionLeader] <defunct>
```

- **Lihat penggunaan memori**

```
danyp@LAPTOP-GRJL2V1D:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       454Mi       3.2Gi       3.4Mi       171Mi       3.3Gi
Swap:          1.0Gi          0B       1.0Gi
```

- **Lihat informasi CPU**

```
danyp@LAPTOP-GRJL2V1D:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   4
  On-line CPU(s) list:    0-3
Vendor ID:                GenuineIntel
  Model name:             11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
    CPU family:           6
    Model:                140
    Thread(s) per core:   2
    Core(s) per socket:   2
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             5990.39
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht sys
                          call nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid
                          tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_ti
                          mer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_s
                          hadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap
                           avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves vnmi avx512vbm
                          i umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rdpid movdiri movdir64b
                           fsrm avx512_vp2intersect md_clear flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    96 KiB (2 instances)
  L1i:                    64 KiB (2 instances)
  L2:                     2.5 MiB (2 instances)
  L3:                     6 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-3
Vulnerabilities:
  Gather data sampling:   Unknown: Dependent on hypervisor status
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Reg file data sampling: Not affected
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBRSB-eIBRS SW sequence; BHI SW loop,
                           KVM SW loop
  Srbds:                  Not affected
  Tsx async abort:        Not affected
danyp@LAPTOP-GRJL2V1D:~$ |
```

- **Lihat perangkat yang terpasang**

```
danyp@LAPTOP-GRJL2V1D:~$ lsblk
NAME
    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.4M  1 disk
sdb   8:16   0   186M  1 disk
sdc   8:32   0     1G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
                               /
```

2.1.2 Program C - Menggunakan Layanan OS

```
danyp@LAPTOP-GRJL2V1D:~$ nano os_services.c
danyp@LAPTOP-GRJL2V1D:~$ gcc -o os_services os_services.c
danyp@LAPTOP-GRJL2V1D:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 575
Parent Process ID (PPID): 287

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
```

2.2 User and Operating-System Interface

User and Operating-System Interface adalah cara pengguna berinteraksi dengan sistem operasi, baik melalui Command Line Interface (CLI) maupun program antarmuka. Dalam praktik Linux ini, pengguna mempelajari perintah dasar seperti `pwd`, `ls`, dan `cd` untuk navigasi, serta fitur pipelining (`|`) dan redirection (`>`, `>>`, `<`) untuk mengalirkan dan mengatur keluaran data. Selain itu, pengguna juga dapat menjalankan proses di latar belakang menggunakan `&`, serta mengelola tugas dengan `jobs` dan `fg`. Program C `simple_shell.c` menunjukkan bagaimana shell sederhana bekerja: membaca input, memecah perintah menjadi argumen, membuat proses baru dengan `fork()`, dan mengeksekusi perintah dengan `execvp()`. Antarmuka ini menjadi jembatan penting antara pengguna dan kernel dalam mengendalikan sistem operasi.

2.2.1 Command Line Interface (CLI)

- **Navigasi dasar**

```
danyp@LAPTOP-GRJL2V1D:~$ pwd
/home/danyp
danyp@LAPTOP-GRJL2V1D:~$ ls -la
total 328
drwxr-x--- 8 danyp danyp  4096 Oct 13 21:09 .
drwxr-xr-x 3 root  root   4096 Oct  7 21:25 ..
-rw------- 1 danyp danyp  8394 Oct 13 20:44 .bash_history
-rw-r--r-- 1 danyp danyp   220 Oct  7 21:25 .bash_logout
-rw-r--r-- 1 danyp danyp  3771 Oct  7 21:25 .bashrc
drwx------ 2 danyp danyp  4096 Oct  7 21:28 .cache
drwx------ 3 danyp danyp  4096 Oct  9 19:58 .config
-rw------- 1 danyp danyp    20 Oct  9 20:10 .lesshst
drwxr-xr-x 3 danyp danyp  4096 Oct  7 21:49 .local
-rw-rw-r-- 1 danyp danyp     0 Oct 13 08:05 .motd_shown
-rw-r--r-- 1 danyp danyp   807 Oct  7 21:25 .profile
drwx------ 2 danyp danyp  4096 Oct  7 23:09 .ssh
-rw-r--r-- 1 danyp danyp     0 Oct  7 22:09 .sudo_as_admin_successful
-rwxr-xr-x 1 danyp danyp 17784 Oct  9 20:24 buggy
-rw-r--r-- 1 danyp danyp   332 Oct  9 20:20 buggy.c
-rw-r--r-- 1 danyp danyp   406 Oct 13 08:34 fifo_reader.c
-rw-r--r-- 1 danyp danyp   508 Oct 13 08:33 fifo_writer.c
-rwxr-xr-x 1 danyp danyp 16224 Oct 13 08:22 fork_demo
-rw-r--r-- 1 danyp danyp   596 Oct 13 08:19 fork_demo.c
-rwxr-xr-x 1 danyp danyp 16680 Oct 13 09:15 ipc_benchmark
-rw-r--r-- 1 danyp danyp  1797 Oct 13 09:24 ipc_benchmark.c
-rwxr-xr-x 1 danyp danyp 17032 Oct  9 20:36 memleak
-rw-r--r-- 1 danyp danyp    95 Oct  9 20:36 memleak.c
-rw-r--r-- 1 danyp danyp   668 Oct 13 08:27 multiple_fork.c
-rwxrwxr-x 1 danyp danyp 16344 Oct  7 23:11 my_daemon
-rw-rw-r-- 1 danyp danyp   729 Oct  7 23:11 my_daemon.c
-rwxr-xr-x 1 danyp danyp 16488 Oct 13 21:09 os_services
-rw-r--r-- 1 danyp danyp  1315 Oct 13 21:09 os_services.c
-rw-r--r-- 1 danyp danyp  1316 Oct  7 21:58 os_services.c:
-rwxr-xr-x 1 danyp danyp 16096 Oct 13 08:09 process_basic
-rw-r--r-- 1 danyp danyp   216 Oct 13 08:09 process_basic.c
-rw-r--r-- 1 danyp danyp   597 Oct 13 08:23 process_wait.c
-rwxr-xr-x 1 danyp danyp 16224 Oct 13 08:34 reader
-rwxr-xr-x 1 danyp danyp 16176 Oct 13 08:45 shm_reader
-rw-r--r-- 1 danyp danyp   576 Oct 13 08:44 shm_reader.c
-rwxr-xr-x 1 danyp danyp 16352 Oct 13 08:45 shm_writer
-rw-r--r-- 1 danyp danyp   690 Oct 13 08:44 shm_writer.c
-rw-r--r-- 1 danyp danyp     0 Oct 13 08:45 shmfile
-rw-r--r-- 1 danyp danyp   898 Oct  7 22:29 simple_shell.c
drwxr-xr-x 2 danyp danyp  4096 Oct 13 19:23 test_dir
-rw-r--r-- 1 danyp danyp    23 Oct 13 21:10 test_file.txt
-rw-r--r-- 1 danyp danyp 10820 Oct  9 20:16 trace.txt
drwxr-xr-x 3 danyp danyp  4096 Oct 11 13:41 tugas_os
-rwxr-xr-x 1 danyp danyp 16352 Oct 13 08:42 writer
danyp@LAPTOP-GRJL2V1D:~$ cd /tmp
danyp@LAPTOP-GRJL2V1D:/tmp$ mkdir test_dir
danyp@LAPTOP-GRJL2V1D:/tmp$ cd test_dir
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Pipelining**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ls -l | grep "^d"
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ps aux | grep bash | wc -l
3
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Redirection**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ echo "Hello World" > output.txt
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat output.txt
Hello World
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat < output.txt
Hello World
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Background**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sleep 30 &
[1] 800
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ jobs
[1]+  Running                 sleep 30 &
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ fg %1
sleep 30
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.2.2 Program C - Shell Sederhana

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano simple_shell.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -o simple_shell simple_shell.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ./simple_shell
myshell> echo Hallo Louis
Hallo Louis
myshell> ^Z
[3]+  Stopped                 ./simple_shell
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.3 System Calls

System calls di Linux adalah mekanisme penting yang menghubungkan program aplikasi dengan kernel. Mereka memungkinkan program untuk melakukan operasi fundamental seperti kontrol proses, manajemen file, dan pemeliharaan informasi sistem. Dalam praktik process control, system calls seperti `fork()` digunakan untuk membuat proses anak, `execlp()` untuk mengeksekusi program baru di proses anak, dan `wait()` agar proses induk menunggu proses anak selesai. Fungsi `getpid()` dan `getppid()` membantu memantau identitas proses, sedangkan file management system calls seperti `open()`, `write()`, `read()`, dan `close()` memungkinkan program membuat, menulis, membaca, dan menutup file. Selain itu, informasi maintenance system calls seperti `getuid()`, `getgid()`, `time()`, dan `times()` membantu memperoleh data pengguna, waktu saat ini, dan statistik eksekusi, mendukung pengelolaan sumber daya sistem secara efisien.

2.3.1 Process Control System Calls

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano process_syscalls.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -o process_syscalls process_syscalls.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 1038
Parent waiting for child (PID: 1039)...
Child PID: 1039
Child's Parent PID: 1038
Child executing 'ls' command...
total 48
-rw-r--r-- 1 danyp danyp    12 Oct 13 21:38 output.txt
-rwxr-xr-x 1 danyp danyp 16360 Oct 13 22:06 process_syscalls
-rw-r--r-- 1 danyp danyp   798 Oct 13 22:06 process_syscalls.c
-rwxr-xr-x 1 danyp danyp 16528 Oct 13 22:05 simple_shell
-rw-r--r-- 1 danyp danyp  1191 Oct 13 21:49 simple_shell.c
Child exited with status: 0
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.3.2 File Management System Calls

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano file_syscalls.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -o file_syscalls file_syscalls.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.3.3 Information Maintenance System Calls

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano info_syscalls.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -o info_syscalls info_syscalls.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 1120
Parent Process ID: 287
User ID: 1000
Group ID: 1000
Current time: Mon Oct 13 22:17:42 2025
Clock ticks since boot: 1718508325
Sleeping for 2 seconds...
Awake!
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.4 System services

System services di Linux adalah program yang berjalan di latar belakang untuk menyediakan fungsi penting bagi sistem dan pengguna. Dengan `systemd`, pengguna dapat mengelola layanan ini secara efisien. Praktik 4.1 menunjukkan cara melihat semua service menggunakan `systemctl list-units --type=service`, memeriksa status service dengan `systemctl status`, dan melihat log terakhir melalui `journalctl`. Layanan yang aktif saat boot dapat ditampilkan dengan `systemctl list-unit-files --state=enabled`. Praktik 4.2 menjelaskan pembuatan custom service, misalnya `my_daemon.c`, yang dijalankan sebagai daemon. Program ini menggunakan `fork()` untuk proses latar belakang, `setsid()` untuk membuat session baru, dan `syslog()` untuk mencatat aktivitas. File service di `/etc/systemd/system/my_daemon.service` mendefinisikan bagaimana daemon dijalankan, dimulai saat boot, dan di-restart jika gagal. Pengelolaan service dilakukan dengan `systemctl daemon-reload`, `start`, `status`, `stop`, dan `journalctl`, sehingga admin dapat memantau dan mengontrol layanan secara aman dan terstruktur.

2.4.1 Menggunakan systemd

- **Lihat semua service**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ systemctl list-units --type=service
  UNIT                                    LOAD   ACTIVE SUB     DESCRIPTION
  console-getty.service                   loaded active running Console Getty
  console-setup.service                   loaded active exited  Set console font and keymap
  cron.service                            loaded active running Regular background program processing daemon
  dbus.service                            loaded active running D-Bus System Message Bus
  getty@tty1.service                      loaded active running Getty on tty1
  keyboard-setup.service                  loaded active exited  Set the console keyboard layout
  kmod-static-nodes.service               loaded active exited  Create List of Static Device Nodes
  rsyslog.service                         loaded active running System Logging Service
  setvtrgb.service                        loaded active exited  Set console scheme
  snapd.seeded.service                    loaded active exited  Wait until snapd is fully seeded
  systemd-binfmt.service                  loaded active exited  Set Up Additional Binary Formats
  systemd-journal-flush.service           loaded active exited  Flush Journal to Persistent Storage
  systemd-journald.service                loaded active running Journal Service
  systemd-logind.service                  loaded active running User Login Management
  systemd-modules-load.service            loaded active exited  Load Kernel Modules
  systemd-remount-fs.service              loaded active exited  Remount Root and Kernel File Systems
  systemd-resolved.service                loaded active running Network Name Resolution
  systemd-sysctl.service                  loaded active exited  Apply Kernel Variables
  systemd-timesyncd.service               loaded active running Network Time Synchronization
  systemd-tmpfiles-setup-dev-early.service loaded active exited  Create Static Device Nodes in /dev gracefully
  systemd-tmpfiles-setup-dev.service      loaded active exited  Create Static Device Nodes in /dev
  systemd-tmpfiles-setup.service          loaded active exited  Create Volatile Files and Directories
  systemd-udev-trigger.service            loaded active exited  Coldplug All udev Devices
  systemd-udevd.service                   loaded active running Rule-based Manager for Device Events and Files
  systemd-update-utmp.service             loaded active exited  Record System Boot/Shutdown in UTMP
  systemd-user-sessions.service           loaded active exited  Permit User Sessions
  unattended-upgrades.service             loaded active running Unattended Upgrades Shutdown
  user-runtime-dir@1000.service           loaded active exited  User Runtime Directory /run/user/1000
  user@1000.service                       loaded active running User Manager for UID 1000
  wsl-pro.service                         loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

30 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
lines 7-38/38 (END)
```

- **Status service**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: inactive (dead)
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat log service**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ journalctl -u ssh -n 20
Oct 07 23:06:42 LAPTOP-GRJL2V1D systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 07 23:06:42 LAPTOP-GRJL2V1D sshd[5606]: Server listening on 0.0.0.0 port 22.
Oct 07 23:06:42 LAPTOP-GRJL2V1D sshd[5606]: Server listening on :: port 22.
Oct 07 23:06:42 LAPTOP-GRJL2V1D systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Oct 07 23:09:32 LAPTOP-GRJL2V1D sshd[5636]: Accepted password for danyp from 172.18.161.244 port 60016 ssh2
Oct 07 23:09:32 LAPTOP-GRJL2V1D sshd[5636]: pam_unix(sshd:session): session opened for user danyp(uid=1000) by danyp(uid=0)
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat service yang berjalan saat boot**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                              STATE   PRESET
apparmor.service                       enabled enabled
apport.service                         enabled enabled
cloud-config.service                   enabled enabled
cloud-final.service                    enabled enabled
cloud-init-local.service               enabled enabled
cloud-init.service                     enabled enabled
console-setup.service                  enabled enabled
cron.service                           enabled enabled
dmesg.service                          enabled enabled
e2scrub_reap.service                   enabled enabled
getty@.service                         enabled enabled
keyboard-setup.service                 enabled enabled
landscape-client.service               enabled enabled
networkd-dispatcher.service            enabled enabled
rsyslog.service                        enabled enabled
setvtrgb.service                       enabled enabled
snapd.apparmor.service                 enabled enabled
snapd.autoimport.service               enabled enabled
snapd.core-fixup.service               enabled enabled
snapd.recovery-chooser-trigger.service enabled enabled
snapd.seeded.service                   enabled enabled
snapd.service                          enabled enabled
snapd.system-shutdown.service          enabled enabled
systemd-pstore.service                 enabled enabled
systemd-resolved.service               enabled enabled
systemd-timesyncd.service              enabled enabled
ua-reboot-cmds.service                 enabled enabled
ubuntu-advantage.service               enabled enabled
unattended-upgrades.service            enabled enabled
wsl-pro.service                        enabled enabled

30 unit files listed.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

2.4.2 Membuat Custom System Service

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano my_daemon.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -o my_daemon my_daemon.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo nano /etc/systemd/system/my_daemon.service
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo systemctl daemon-reload
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo systemctl start my_daemon
Job for my_daemon.service failed because the control process exited with error code.
See "systemctl status my_daemon.service" and "journalctl -xeu my_daemon.service" for details.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo systemctl status my_daemon
× my_daemon.service - My Custom Daemon
     Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
     Active: failed (Result: exit-code) since Mon 2025-10-13 22:50:04 WIB; 8s ago
    Process: 1472 ExecStart=/path/to/my_daemon (code=exited, status=203/EXEC)
        CPU: 2ms

Oct 13 22:50:04 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 13 22:50:04 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 13 22:50:04 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 13 22:50:04 LAPTOP-GRJL2V1D systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo journalctl -u my_daemon
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 14:52:06 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 4.
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 14:52:07 LAPTOP-GRJL2V1D systemd[1]: Failed to start my_daemon.service - My Simple Daemon.
Oct 09 14:57:38 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:57:38 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Deactivated successfully.
Oct 09 14:57:39 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 14:57:39 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:57:39 LAPTOP-GRJL2V1D my_daemon[1045]: My daemon started
Oct 09 14:57:39 LAPTOP-GRJL2V1D my_daemon[1045]: Daemon is running...
Oct 09 14:57:39 LAPTOP-GRJL2V1D my_daemon[1045]: My daemon stopped
Oct 09 14:57:39 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Deactivated successfully.
Oct 09 14:57:39 LAPTOP-GRJL2V1D systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 14:57:39 LAPTOP-GRJL2V1D systemd[1]: Started my_daemon.service - My Simple Daemon.
Oct 09 14:57:39 LAPTOP-GRJL2V1D my_daemon[1048]: My daemon started
Oct 09 14:57:39 LAPTOP-GRJL2V1D my_daemon[1048]: Daemon is running...
lines 1-35
[7]+  Stopped                 sudo journalctl -u my_daemon
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo systemctl stop my_daemon
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

2.5 Linkers and Loaders

Linkers dan loaders di Linux adalah komponen penting dalam proses eksekusi program. Linker bertugas menggabungkan berbagai file objek hasil kompilasi menjadi satu executable, menyelesaikan referensi simbol dan alamat fungsi atau variabel. Praktik 5.1 menunjukkan proses linking bertahap: preprocessing menghasilkan file `.i`, compilation menghasilkan file assembly `.s`, assembly menjadi object file `.o`, lalu linking menghasilkan executable. Perintah `nm` menampilkan simbol, sedangkan `ldd` menampilkan dependensi library. Praktik 5.2 membandingkan static dan dynamic linking, di mana static menggabungkan library ke executable sehingga ukuran lebih besar, sedangkan dynamic tetap bergantung pada library eksternal saat runtime. Praktik 5.3 menunjukkan pembuatan shared library `.so`, kompilasi program yang menggunakannya, dan penggunaan `LD_LIBRARY_PATH` untuk menjalankan program. Linkers dan loaders memastikan kode dapat dijalankan dengan benar, efisien, dan terstruktur, mendukung modularitas dan pemeliharaan program di Linux.

2.5.1 Proses Linking

- **Buat file math_ops.h**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano math_ops.h
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Buat file math_ops.c**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano math_ops.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Buat main.c**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano main.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Preprocessing**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -E main.c -o main.i
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -E math_ops.c -o math_ops.i
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Compilation (assembly)**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -S main.c -o main.s
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -S math_ops.c -o math_ops.s
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Assembly (object files)**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -c main.c -o main.o
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -c math_ops.c -o math_ops.o
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Linking**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc main.o math_ops.o -o program
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat symbols dalam object file**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nm main.o
                 U add
0000000000000000 T main
                 U multiply
                 U printf
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
                 w _ITM_deregisterTMCloneTable
                 w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
0000000000004010 B __bss_start
                 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
                 w __gmon_start__
                 U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
                 U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat dependencies**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ldd program
        linux-vdso.so.1 (0x00007fffbbf55000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000074e7cbc00000)
        /lib64/ld-linux-x86-64.so.2 (0x000074e7cbf08000)
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.5.2 Static vs Dynamic Linking

- **Static linking**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc main.c math_ops.c -o program_static -static
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Dynamic linking**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc main.c math_ops.c -o program_dynamic
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Bandingkan ukuran**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 danyp danyp  16K Oct 13 23:14 program_dynamic
-rwxr-xr-x 1 danyp danyp 767K Oct 13 23:14 program_static
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat dependencies**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ldd program_static
        not a dynamic executable
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ldd program_dynamic
        linux-vdso.so.1 (0x00007fff4089e000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007d260dc00000)
        /lib64/ld-linux-x86-64.so.2 (0x00007d260dff7000)
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.5.3 Membuat dan Menggunakan Shared Library

- **Compile sebagai shared library**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -c -fPIC math_ops.c -o math_ops.o
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -shared -o libmathops.so math_ops.o
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Compile program yang menggunakan library**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc main.c -L. -lmathops -o program_shared
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Jalankan dengan LD_LIBRARY_PATH**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat informasi loader**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
  Tag        Type                         Name/Value
 0x0000000000000001 (NEEDED)             Shared library: [libmathops.so]
 0x0000000000000001 (NEEDED)             Shared library: [libc.so.6]
 0x000000000000000c (INIT)               0x1000
 0x000000000000000d (FINI)               0x11e4
 0x0000000000000019 (INIT_ARRAY)         0x3d98
 0x000000000000001b (INIT_ARRAYSZ)       8 (bytes)
 0x000000000000001a (FINI_ARRAY)         0x3da0
 0x000000000000001c (FINI_ARRAYSZ)       8 (bytes)
 0x000000006ffffef5 (GNU_HASH)           0x3b0
 0x0000000000000005 (STRTAB)             0x4b0
 0x0000000000000006 (SYMTAB)             0x3d8
 0x000000000000000a (STRSZ)              170 (bytes)
 0x000000000000000b (SYMENT)             24 (bytes)
 0x0000000000000015 (DEBUG)              0x0
 0x0000000000000003 (PLTGOT)             0x3fa8
 0x0000000000000002 (PLTRELSZ)           72 (bytes)
 0x0000000000000014 (PLTREL)             RELA
 0x0000000000000017 (JMPREL)             0x660
 0x0000000000000007 (RELA)               0x5a0
 0x0000000000000008 (RELASZ)             192 (bytes)
 0x0000000000000009 (RELAENT)            24 (bytes)
 0x000000000000001e (FLAGS)              BIND_NOW
 0x000000006ffffffb (FLAGS_1)            Flags: NOW PIE
 0x000000006ffffffe (VERNEED)            0x570
 0x000000006fffffff (VERNEEDNUM)         1
 0x000000006ffffff0 (VERSYM)             0x55a
 0x000000006ffffff9 (RELACOUNT)          3
 0x0000000000000000 (NULL)               0x0
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.6 Why Applications Are Operating-System Specific

Aplikasi bersifat spesifik terhadap sistem operasi karena setiap OS menyediakan set system call, API, dan format file eksekusi yang berbeda. Praktik 6.1 menunjukkan perbedaan system call antara Linux dan Windows, di mana Linux menggunakan header seperti `<unistd.h>` dan fungsi `uname()` atau `getpid()`, sementara Windows menggunakan `<windows.h>` dan API khususnya. Hal ini membuat kode yang sama harus dimodifikasi agar berjalan di OS berbeda. Praktik 6.2 menekankan analisis format biner, misalnya executable Linux menggunakan format ELF, sementara Windows menggunakan PE. Perintah seperti `readelf` dan `objdump` memungkinkan memeriksa header, section, program headers, dan disassembly. Perbedaan ini menjelaskan mengapa aplikasi tidak bisa dijalankan secara langsung di OS lain tanpa porting atau emulasi, karena perbedaan ABI, manajemen memori, dan dukungan library bawaan.

2.6.1 System Call Differences

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano os_specific.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.6.2 Binary Format Analysis

- **Lihat format executable**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildI
D[sha1]=0c5145e0b63b47095af229368edc342f23076fc2, for GNU/Linux 3.2.0, not stripped
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat header ELF**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:                             ELF64
  Data:                              2's complement, little endian
  Version:                           1 (current)
  OS/ABI:                            UNIX - System V
  ABI Version:                       0
  Type:                              DYN (Position-Independent Executable file)
  Machine:                           Advanced Micro Devices X86-64
  Version:                           0x1
  Entry point address:               0x1060
  Start of program headers:          64 (bytes into file)
  Start of section headers:          14072 (bytes into file)
  Flags:                             0x0
  Size of this header:               64 (bytes)
  Size of program headers:           56 (bytes)
  Number of program headers:         13
  Size of section headers:           64 (bytes)
  Number of section headers:         31
  Section header string table index: 30
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat section haders**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:

Section Headers:
  [Nr] Name              Type             Address           Offset
       Size              EntSize          Flags  Link  Info  Align
  [ 0]                   NULL             0000000000000000  00000000
       0000000000000000  0000000000000000           0     0     0
  [ 1] .interp           PROGBITS         0000000000000318  00000318
       000000000000001c  0000000000000000   A       0     0     1
  [ 2] .note.gnu.pr[...] NOTE             0000000000000338  00000338
       0000000000000030  0000000000000000   A       0     0     8
  [ 3] .note.gnu.bu[...] NOTE             0000000000000368  00000368
       0000000000000024  0000000000000000   A       0     0     4
  [ 4] .note.ABI-tag     NOTE             000000000000038c  0000038c
       0000000000000020  0000000000000000   A       0     0     4
  [ 5] .gnu.hash         GNU_HASH         00000000000003b0  000003b0
       0000000000000024  0000000000000000   A       6     0     8
  [ 6] .dynsym           DYNSYM           00000000000003d8  000003d8
       00000000000000a8  0000000000000018   A       7     1     8
  [ 7] .dynstr           STRTAB           0000000000000480  00000480
       000000000000008f  0000000000000000   A       0     0     1
  [ 8] .gnu.version      VERSYM           0000000000000510  00000510
       000000000000000e  0000000000000002   A       6     0     2
  [ 9] .gnu.version_r    VERNEED          0000000000000520  00000520
       0000000000000030  0000000000000000   A       7     1     8
  [10] .rela.dyn         RELA             0000000000000550  00000550
       00000000000000c0  0000000000000018   A       6     0     8
  [11] .rela.plt         RELA             0000000000000610  00000610
       0000000000000018  0000000000000018  AI       6    24     8
  [12] .init             PROGBITS         0000000000001000  00001000
       000000000000001b  0000000000000000  AX       0     0     4
  [13] .plt              PROGBITS         0000000000001020  00001020
       0000000000000020  0000000000000010  AX       0     0     16
  [14] .plt.got          PROGBITS         0000000000001040  00001040
       0000000000000010  0000000000000010  AX       0     0     16
```

```
  [15] .plt.sec          PROGBITS         0000000000001050  00001050
       0000000000000010  0000000000000010  AX       0     0     16
  [16] .text             PROGBITS         0000000000001060  00001060
       0000000000000171  0000000000000000  AX       0     0     16
  [17] .fini             PROGBITS         00000000000011d4  000011d4
       000000000000000d  0000000000000000  AX       0     0     4
  [18] .rodata           PROGBITS         0000000000002000  00002000
       000000000000001c  0000000000000000   A       0     0     4
  [19] .eh_frame_hdr     PROGBITS         000000000000201c  0000201c
       0000000000000044  0000000000000000   A       0     0     4
  [20] .eh_frame         PROGBITS         0000000000002060  00002060
       00000000000000ec  0000000000000000   A       0     0     8
  [21] .init_array       INIT_ARRAY       0000000000003db8  00002db8
       0000000000000008  0000000000000008  WA       0     0     8
  [22] .fini_array       FINI_ARRAY       0000000000003dc0  00002dc0
       0000000000000008  0000000000000008  WA       0     0     8
  [23] .dynamic          DYNAMIC          0000000000003dc8  00002dc8
       00000000000001f0  0000000000000010  WA       7     0     8
  [24] .got              PROGBITS         0000000000003fb8  00002fb8
       0000000000000048  0000000000000008  WA       0     0     8
  [25] .data             PROGBITS         0000000000004000  00003000
       0000000000000010  0000000000000000  WA       0     0     8
  [26] .bss              NOBITS           0000000000004010  00003010
       0000000000000008  0000000000000000  WA       0     0     1
  [27] .comment          PROGBITS         0000000000000000  00003010
       000000000000002b  0000000000000001  MS       0     0     1
  [28] .symtab           SYMTAB           0000000000000000  00003040
       00000000000003a8  0000000000000018          29    19     8
  [29] .strtab           STRTAB           0000000000000000  000033e8
       00000000000001f4  0000000000000000           0     0     1
  [30] .shstrtab         STRTAB           0000000000000000  000035dc
       000000000000011a  0000000000000000           0     0     1
Key to Flags:
  W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
  L (link order), O (extra OS processing required), G (group), T (TLS),
  C (compressed), x (unknown), o (OS specific), E (exclude),
  D (mbind), l (large), p (processor specific)
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat program headers**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ readelf -l program

Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1060
There are 13 program headers, starting at offset 64

Program Headers:
  Type           Offset             VirtAddr           PhysAddr
                 FileSiz            MemSiz              Flags  Align
  PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
                 0x00000000000002d8 0x00000000000002d8  R      0x8
  INTERP         0x0000000000000318 0x0000000000000318 0x0000000000000318
                 0x000000000000001c 0x000000000000001c  R      0x1
      [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
  LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
                 0x0000000000000628 0x0000000000000628  R      0x1000
  LOAD           0x0000000000001000 0x0000000000001000 0x0000000000001000
                 0x00000000000001e1 0x00000000000001e1  R E    0x1000
  LOAD           0x0000000000002000 0x0000000000002000 0x0000000000002000
                 0x000000000000014c 0x000000000000014c  R      0x1000
  LOAD           0x0000000000002db8 0x0000000000003db8 0x0000000000003db8
                 0x0000000000000258 0x0000000000000260  RW     0x1000
  DYNAMIC        0x0000000000002dc8 0x0000000000003dc8 0x0000000000003dc8
                 0x00000000000001f0 0x00000000000001f0  RW     0x8
  NOTE           0x0000000000000338 0x0000000000000338 0x0000000000000338
                 0x0000000000000030 0x0000000000000030  R      0x8
  NOTE           0x0000000000000368 0x0000000000000368 0x0000000000000368
                 0x0000000000000044 0x0000000000000044  R      0x4
  GNU_PROPERTY   0x0000000000000338 0x0000000000000338 0x0000000000000338
                 0x0000000000000030 0x0000000000000030  R      0x8
  GNU_EH_FRAME   0x000000000000201c 0x000000000000201c 0x000000000000201c
                 0x0000000000000044 0x0000000000000044  R      0x4
  GNU_STACK      0x0000000000000000 0x0000000000000000 0x0000000000000000
                 0x0000000000000000 0x0000000000000000  RW     0x10
  GNU_RELRO      0x0000000000002db8 0x0000000000003db8 0x0000000000003db8
                 0x0000000000000248 0x0000000000000248  R      0x1

 Section to Segment mapping:
  Segment Sections...
   00
   01     .interp
   02     .interp .note.gnu.property .note.gnu.build-id .note.ABI-tag .gnu.hash .dynsym .dynstr .gnu.version .gnu.version_r .rela.dyn .rela.plt
   03     .init .plt .plt.got .plt.sec .text .fini
   04     .rodata .eh_frame_hdr .eh_frame
   05     .init_array .fini_array .dynamic .got .data .bss
   06     .dynamic
   07     .note.gnu.property
   08     .note.gnu.build-id .note.ABI-tag
   09     .note.gnu.property
   10     .eh_frame_hdr
   11
   12     .init_array .fini_array .dynamic .got
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Disassemble**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ objdump -d program | head -50

program:     file format elf64-x86-64


Disassembly of section .init:

0000000000001000 <_init>:
    1000:       f3 0f 1e fa             endbr64
    1004:       48 83 ec 08             sub    $0x8,%rsp
    1008:       48 8b 05 d9 2f 00 00    mov    0x2fd9(%rip),%rax        # 3fe8 <__gmon_start__@Base>
    100f:       48 85 c0                test   %rax,%rax
    1012:       74 02                   je     1016 <_init+0x16>
    1014:       ff d0                   call   *%rax
    1016:       48 83 c4 08             add    $0x8,%rsp
    101a:       c3                      ret

Disassembly of section .plt:

0000000000001020 <.plt>:
    1020:       ff 35 9a 2f 00 00       push   0x2f9a(%rip)        # 3fc0 <_GLOBAL_OFFSET_TABLE_+0x8>
    1026:       ff 25 9c 2f 00 00       jmp    *0x2f9c(%rip)        # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>
    102c:       0f 1f 40 00             nopl   0x0(%rax)
    1030:       f3 0f 1e fa             endbr64
    1034:       68 00 00 00 00          push   $0x0
    1039:       e9 e2 ff ff ff          jmp    1020 <_init+0x20>
    103e:       66 90                   xchg   %ax,%ax

Disassembly of section .plt.got:

0000000000001040 <__cxa_finalize@plt>:
    1040:       f3 0f 1e fa             endbr64
    1044:       ff 25 ae 2f 00 00       jmp    *0x2fae(%rip)        # 3ff8 <__cxa_finalize@GLIBC_2.2.5>
    104a:       66 0f 1f 44 00 00       nopw   0x0(%rax,%rax,1)

Disassembly of section .plt.sec:

0000000000001050 <printf@plt>:
    1050:       f3 0f 1e fa             endbr64
    1054:       ff 25 76 2f 00 00       jmp    *0x2f76(%rip)        # 3fd0 <printf@GLIBC_2.2.5>
    105a:       66 0f 1f 44 00 00       nopw   0x0(%rax,%rax,1)

Disassembly of section .text:

0000000000001060 <_start>:
    1060:       f3 0f 1e fa             endbr64
    1064:       31 ed                   xor    %ebp,%ebp
    1066:       49 89 d1                mov    %rdx,%r9
    1069:       5e                      pop    %rsi
    106a:       48 89 e2                mov    %rsp,%rdx
    106d:       48 83 e4 f0             and    $0xfffffffffffffff0,%rsp
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

2.7 OS Design and Implementation

Desain dan implementasi sistem operasi Linux melibatkan kernel sebagai inti yang mengatur sumber daya perangkat keras, manajemen proses, dan interaksi dengan aplikasi. Praktik 7.1 menunjukkan pembuatan kernel module sederhana hello_module.c, yang dapat dimuat ke kernel menggunakan insmod dan dihapus dengan rmmod. Kernel module memungkinkan penambahan fungsi ke kernel tanpa mengubah kode inti, dicatat melalui printk() dan dipantau dengan dmesg. Makefile digunakan untuk mengompilasi modul terhadap build kernel saat ini. Praktik 7.2 menekankan monitoring kernel: uname -r dan /proc/version menampilkan versi kernel, sysctl -a melihat parameter kernel, lsmod menampilkan modul yang dimuat, dan modinfo memberikan detail modul tertentu. Pemahaman modul dan monitoring kernel penting untuk pengembangan OS, debugging, dan optimasi performa, menunjukkan bagaimana desain modular dan monitoring internal memungkinkan Linux fleksibel dan dapat diperluas secara aman.

2.7.1 Kernel Module Sederhana

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano hello_module.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Makefile**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo nano Makefile
```

- **Compile dan load**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ make
make -C /lib/modules/6.6.87.2-microsoft-standard-WSL2/build M=/tmp/test_dir modules
make[1]: *** /lib/modules/6.6.87.2-microsoft-standard-WSL2/build: No such file or directory.  Stop.
make: *** [Makefile:4: all] Error 2
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ lsmod | grep hello
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg | tail
[   48.496804] hv_balloon: Max. dynamic memory size: 3986 MB
[  600.238339] mini_init (174): drop_caches: 1
[ 5692.811665] systemd-journald[43]: Time jumped backwards, rotating.
[ 7296.573727] systemd-journald[43]: Time jumped backwards, rotating.
[ 7487.364857] systemd-journald[43]: Time jumped backwards, rotating.
[ 7900.199291] systemd-journald[43]: Time jumped backwards, rotating.
[ 8470.174075] systemd-journald[43]: Time jumped backwards, rotating.
[11031.730693] Invalid ELF header magic: != \x7fELF
[11614.260450] systemd-journald[43]: Time jumped backwards, rotating.
[11827.851284] Invalid ELF header magic: != \x7fELF
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg | tail
[   48.496804] hv_balloon: Max. dynamic memory size: 3986 MB
[  600.238339] mini_init (174): drop_caches: 1
[ 5692.811665] systemd-journald[43]: Time jumped backwards, rotating.
[ 7296.573727] systemd-journald[43]: Time jumped backwards, rotating.
[ 7487.364857] systemd-journald[43]: Time jumped backwards, rotating.
[ 7900.199291] systemd-journald[43]: Time jumped backwards, rotating.
[ 8470.174075] systemd-journald[43]: Time jumped backwards, rotating.
[11031.730693] Invalid ELF header magic: != \x7fELF
[11614.260450] systemd-journald[43]: Time jumped backwards, rotating.
[11827.851284] Invalid ELF header magic: != \x7fELF
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.7.2 Monitoring Kernel

- **Lihat informasi kernel**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ uname -r
6.6.87.2-microsoft-standard-WSL2
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNA
MIC Thu Jun  5 18:30:46 UTC 2025
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat parameter kernel**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'abi.vsyscall32 = 1

debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat kernel ring buffer**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg | tail -30
[    1.428113] sd 0:0:0:3: [sdd] Mode Sense: 0f 00 00 00
[    1.428423] sd 0:0:0:3: [sdd] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
[    1.429525] sd 0:0:0:3: [sdd] Attached SCSI disk
[    1.446609] EXT4-fs (sdd): mounted filesystem f770c218-c434-4baf-b33c-877731a81b22 r/w with ordered data mode. Quota mode: none.
[    1.994086] pulseaudio[220]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[    2.142989] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[    2.187696] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[    2.189786] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[    2.191979] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[    2.193889] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[    2.270177] Failed to connect to bus: No such file or directory
[    2.564605] systemd-journald[43]: Collecting audit messages is disabled.
[    2.843793] systemd-journald[43]: Received client request to flush runtime journal.
[    2.847600] systemd-journald[43]: File /var/log/journal/1cb2442bdf224e8691d1e1948ea146fb/system.journal corrupted or uncleanly shu
t down, renaming and replacing.
[    3.305748] ACPI: battery: Slot [BAT1] (battery present)
[    3.311244] ACPI: AC: AC Adapter [AC1] (on-line)
[    4.397421] kvm_intel: Using Hyper-V Enlightened VMCS
[    4.617344] intel_rapl_msr: PL4 support detected.
[   11.224293] WSL (188) ERROR: CheckConnection: getaddrinfo() failed: -3
[   48.496804] hv_balloon: Max. dynamic memory size: 3986 MB
[  600.238339] mini_init (174): drop_caches: 1
[ 5692.811665] systemd-journald[43]: Time jumped backwards, rotating.
[ 7296.573727] systemd-journald[43]: Time jumped backwards, rotating.
[ 7487.364857] systemd-journald[43]: Time jumped backwards, rotating.
[ 7900.199291] systemd-journald[43]: Time jumped backwards, rotating.
[ 8470.174075] systemd-journald[43]: Time jumped backwards, rotating.
[11031.730693] Invalid ELF header magic: != \x7fELF
[11614.260450] systemd-journald[43]: Time jumped backwards, rotating.
[11827.851284] Invalid ELF header magic: != \x7fELF
[12073.334460] systemd-journald[43]: Time jumped backwards, rotating.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat kernel modules**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ lsmod
Module                  Size  Used by
tls                    98304  0
intel_rapl_msr         16384  0
intel_rapl_common      32768  1 intel_rapl_msr
kvm_intel             356352  0
kvm                   970752  1 kvm_intel
irqbypass              12288  1 kvm
ac                     16384  0
crc32c_intel           16384  0
battery                20480  0
sch_fq_codel           16384  1
configfs               53248  1
autofs4                45056  0
br_netfilter           28672  0
bridge                282624  1 br_netfilter
stp                    12288  1 bridge
llc                    12288  2 bridge,stp
ip_tables              28672  0
tun                    53248  0
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Informasi modul tertentu**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ modinfo ext4
name:           ext4
filename:       (builtin)
softdep:        pre: crc32c
license:        GPL
file:           fs/ext4/ext4
description:    Fourth Extended Filesystem
author:         Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias:          fs-ext4
alias:          ext3
alias:          fs-ext3
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.8 Operating-System Structure

Struktur sistem operasi Linux terdiri dari beberapa lapisan yang saling berinteraksi, mulai dari perangkat keras hingga aplikasi pengguna. Praktik 8.1 memperlihatkan konsep layering, di mana Layer 1 (Hardware) menampilkan perangkat fisik melalui perintah seperti lspci, lsusb, dan cat /proc/cpuinfo. Layer 2 (Kernel) berfungsi mengatur komunikasi dengan perangkat keras menggunakan uname -a dan cat /proc/sys/kernel/ostype. Layer 3 (System Calls) memperlihatkan interaksi antara aplikasi dan kernel melalui perintah strace, sedangkan Layer 4 (System Programs) menjalankan program manajemen sistem seperti ps aux dan top. Pada Layer 5 (User Applications), terdapat aplikasi pengguna seperti bash dan gcc. Praktik 8.2 membahas perbedaan antara monolithic kernel dan microkernel, di mana Linux menggunakan kernel monolitik yang menggabungkan banyak fungsi dalam satu kesatuan besar, terlihat dari ukuran file /boot/vmlinuz. Sementara itu, Praktik 8.3 menunjukkan penggunaan Virtual File System melalui program procfs_read.c yang membaca informasi CPU, memori, dan proses aktif dari direktori /proc. Struktur berlapis ini menggambarkan bagaimana sistem operasi Linux mengelola sumber daya secara efisien, menjaga stabilitas, dan memberikan akses terorganisir antara perangkat keras dan aplikasi pengguna.

2.8.1 Eksplorasi Struktur Layering

- **Layer 1: Hardware**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ lspci
5582:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
64df:00:00.0 System peripheral: Red Hat, Inc. Virtio file system (rev 01)
d42b:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ lsusb
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 140
model name      : 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
stepping        : 2
microcode       : 0xffffffff
cpu MHz         : 2995.198
cache size      : 6144 KB
physical id     : 0
siblings        : 4
core id         : 0
cpu cores       : 2
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 27
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm const
ant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic m
ovbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad
fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt
xsavec xgetbv1 xsaves vnmi avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rdpid movdiri movdir64b fsrm avx512_
vp2intersect md_clear flush_l1d arch_capabilities
vmx flags       : vnmi invvpid ept_x_only ept_ad ept_1gb tsc_offset vtpr ept vpid unrestricted_guest ept_mode_based_exec tsc_scaling
bugs            : spectre_v1 spectre_v2 spec_store_bypass swapgs retbleed eibrs_pbrsb gds bhi
bogomips        : 5990.39
clflush size    : 64
cache_alignment : 64
address sizes   : 39 bits physical, 48 bits virtual
power management:
```

- **Layer 2: Kernel**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ uname -a
Linux LAPTOP-GRJL2V1D 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat /proc/sys/kernel/ostype
Linux
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Layer 3: System Calls**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffd1a1da280 /* 27 vars */) = 0
brk(NULL)                               = 0x577e12a02000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7e354d562000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20143, ...}) = 0
mmap(NULL, 20143, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7e354d55d000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7e354d530000
mmap(0x7e354d536000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x7e354d536000
mmap(0x7e354d553000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x7e354d553000
mmap(0x7e354d559000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x7e354d559000
mmap(0x7e354d55b000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7e354d55b000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7e354d200000
mmap(0x7e354d228000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7e354d228000
mmap(0x7e354d3b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x7e354d3b0000
mmap(0x7e354d3ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x7e354d3ff000
mmap(0x7e354d405000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7e354d405000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7e354d496000
mmap(0x7e354d498000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7e354d498000
mmap(0x7e354d506000, 163840, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x7e354d506000
mmap(0x7e354d52e000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x97000) = 0x7e354d52e000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7e354d493000
arch_prctl(ARCH_SET_FS, 0x7e354d493800) = 0
set_tid_address(0x7e354d493ad0)         = 2355
```

- **Layer 4: System Programs**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  22184 13032 ?        Ss   Oct13   0:03 /sbin/init
root           2  0.0  0.0   3072  1664 ?        Sl   Oct13   0:00 /init
root           6  0.0  0.0   3072  1792 ?        Sl   Oct13   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          43  0.0  0.4  66828 19020 ?        S<s  Oct13   0:03 /usr/lib/systemd/systemd-journald
root          97  0.0  0.1  25536  6656 ?        Ss   Oct13   0:06 /usr/lib/systemd/systemd-udevd
systemd+     108  0.0  0.3  21456 12672 ?        Ss   Oct13   0:00 /usr/lib/systemd/systemd-resolved
systemd+     112  0.0  0.2  91024  7808 ?        Ssl  Oct13   0:00 /usr/lib/systemd/systemd-timesyncd
root         161  0.0  0.0   4236  2560 ?        Ss   Oct13   0:00 /usr/sbin/cron -f -P
message+     162  0.0  0.1   9628  5120 ?        Ss   Oct13   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
root         169  0.0  0.2  17968  8576 ?        Ss   Oct13   0:00 /usr/lib/systemd/systemd-logind
root         182  0.0  0.3 1756096 13824 ?       Ssl  Oct13   0:01 /usr/libexec/wsl-pro-service -vv
syslog       185  0.0  0.1 222508  5504 ?        Ssl  Oct13   0:00 /usr/sbin/rsyslogd -n -iNONE
root         189  0.0  0.0   3160  1792 hvc0     Ss+  Oct13   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root         193  0.0  0.0   3116  1920 tty1     Ss+  Oct13   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root         198  0.0  0.5 107012 22528 ?        Ssl  Oct13   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
root         285  0.0  0.0   3076   896 ?        Ss   Oct13   0:00 /init
root         286  0.0  0.0   3092  1152 ?        S    Oct13   0:01 /init
danyp        287  0.0  0.1   6204  5248 pts/0    Ss   Oct13   0:00 -bash
root         288  0.0  0.0      0     0 ?        Zs   Oct13   0:00 [SessionLeader] <defunct>
root         290  0.0  0.1   6820  4352 pts/1    Ss   Oct13   0:00 /bin/login -f
danyp        334  0.0  0.2  20308 11136 ?        Ss   Oct13   0:00 /usr/lib/systemd/systemd --user
danyp        335  0.0  0.0  21148  3516 ?        S    Oct13   0:00 (sd-pam)
danyp        354  0.0  0.1   6072  4864 pts/1    S+   Oct13   0:00 -bash
danyp        856  0.0  0.0   2680  1536 pts/0    T    Oct13   0:00 ./simple_shell
danyp        872  0.0  0.0   2680  1536 pts/0    T    Oct13   0:00 ./simple_shell
danyp        885  0.0  0.0   2680  1536 pts/0    T    Oct13   0:00 ./simple_shell
danyp       1025  0.0  0.0   2680  1280 pts/0    T    Oct13   0:00 ./simple_shell
danyp       1248  0.0  0.1  14772  6144 pts/0    T    Oct13   0:00 systemctl list-units --type=service
danyp       1249  0.0  0.0   3512  2304 pts/0    T    Oct13   0:00 less
danyp       1315  0.0  0.0   2680  1152 ?        Ss   Oct13   0:00 ./my_daemon
root        1409  0.0  0.1  14148  6912 pts/0    T    Oct13   0:00 sudo journalctl -u my_daemon
root        1410  0.0  0.0  14148  2600 pts/2    Ss   Oct13   0:00 sudo journalctl -u my_daemon
root        1411  0.0  0.1  17876  7676 pts/2    T+   Oct13   0:00 journalctl -u my_daemon
root        1412  0.0  0.0   3512  2304 pts/2    S+   Oct13   0:00 less
root        1480  0.0  0.1  14148  6912 pts/0    T    Oct13   0:00 sudo journalctl -u my_daemon
root        1481  0.0  0.0  14148  2600 pts/3    Ss   Oct13   0:00 sudo journalctl -u my_daemon
root        1482  0.0  0.2  17876  7808 pts/3    T+   Oct13   0:00 journalctl -u my_daemon
root        1483  0.0  0.0   3512  2304 pts/3    S+   Oct13   0:00 less
danyp       2362  300  0.1   8332  4224 pts/0    R+   00:33   0:00 ps aux
```

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ top
top - 00:33:21 up  3:49,  1 user,  load average: 0.02, 0.01, 0.00
Tasks:  39 total,   1 running,  27 sleeping,  10 stopped,   1 zombie
%Cpu(s):  0.1 us,  0.1 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3807.2 total,   3110.5 free,    468.7 used,    364.9 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3338.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  286 root      20   0    3092   1152   1024 S   0.3   0.0   0:01.27 Relay(287)
 2365 danyp     20   0    9396   5632   3456 R   0.3   0.1   0:00.03 top
    1 root      20   0   22184  13032   9448 S   0.0   0.3   0:03.79 systemd
    2 root      20   0    3072   1664   1664 S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root      20   0    3072   1792   1792 S   0.0   0.0   0:00.00 init
   43 root      19  -1   66828  19020  18124 S   0.0   0.5   0:03.12 systemd-journal
   97 root      20   0   25536   6656   4992 S   0.0   0.2   0:06.32 systemd-udevd
  108 systemd+  20   0   21456  12672  10496 S   0.0   0.3   0:00.39 systemd-resolve
  112 systemd+  20   0   91024   7808   6912 S   0.0   0.2   0:00.67 systemd-timesyn
  161 root      20   0    4236   2560   2432 S   0.0   0.1   0:00.06 cron
  162 message+  20   0    9628   5120   4480 S   0.0   0.1   0:00.91 dbus-daemon
  169 root      20   0   17968   8576   7680 S   0.0   0.2   0:00.45 systemd-logind
  182 root      20   0 1756096  13824  10624 S   0.0   0.4   0:01.35 wsl-pro-service
  185 syslog    20   0  222508   5504   4352 S   0.0   0.1   0:00.51 rsyslogd
  189 root      20   0    3160   1792   1792 S   0.0   0.0   0:00.00 agetty
  193 root      20   0    3116   1920   1792 S   0.0   0.0   0:00.00 agetty
  198 root      20   0  107012  22528  13184 S   0.0   0.6   0:00.24 unattended-upgr
  285 root      20   0    3076    896    896 S   0.0   0.0   0:00.00 SessionLeader
  287 danyp     20   0    6204   5248   3584 S   0.0   0.1   0:00.64 bash
  288 root      20   0       0      0      0 Z   0.0   0.0   0:00.00 SessionLeader
  290 root      20   0    6820   4352   3840 S   0.0   0.1   0:00.03 login
  334 danyp     20   0   20308  11136   9216 S   0.0   0.3   0:00.24 systemd
  335 danyp     20   0   21148   3516   1792 S   0.0   0.1   0:00.00 (sd-pam)
  354 danyp     20   0    6072   4864   3328 S   0.0   0.1   0:00.01 bash
  856 danyp     20   0    2680   1536   1536 T   0.0   0.0   0:00.00 simple_shell
  872 danyp     20   0    2680   1536   1536 T   0.0   0.0   0:00.00 simple_shell
  885 danyp     20   0    2680   1536   1536 T   0.0   0.0   0:00.00 simple_shell
 1025 danyp     20   0    2680   1280   1280 T   0.0   0.0   0:00.00 simple_shell
 1248 danyp     20   0   14772   6144   5376 T   0.0   0.2   0:00.06 systemctl
 1249 danyp     20   0    3512   2304   2048 T   0.0   0.1   0:00.01 less
 1315 danyp     20   0    2680   1152   1152 S   0.0   0.0   0:00.07 my_daemon
 1409 root      20   0   14148   6912   5760 T   0.0   0.2   0:00.01 sudo
 1410 root      20   0   14148   2600   1408 S   0.0   0.1   0:00.00 sudo
 1411 root      20   0   17876   7676   6780 T   0.0   0.2   0:00.13 journalctl
 1412 root      20   0    3512   2304   2048 S   0.0   0.1   0:00.01 less
 1480 root      20   0   14148   6912   5760 T   0.0   0.2   0:00.00 sudo
 1481 root      20   0   14148   2600   1408 S   0.0   0.1   0:00.00 sudo
 1482 root      20   0   17876   7808   6912 T   0.0   0.2   0:00.01 journalctl
 1483 root      20   0    3512   2304   2048 S   0.0   0.1   0:00.00 less
```

- **Layer 5: User Applications**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$  which bash
/usr/bin/bash
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ which gcc
/usr/bin/gcc
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.8.2 Microkernel vs Monolithic

- **Lihat ukuran kernel (Monolithic)**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat built-in vs module**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Lihat loaded modules**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ lsmod | wc -l
19
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.8.3 Program - Virtual File System

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano procfs_read.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -o procfs_read procfs_read.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ./procfs_read
=== CPU Information ===
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 140
model name      : 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
stepping        : 2
microcode       : 0xffffffff
cpu MHz         : 2995.198
cache size      : 6144 KB
physical id     : 0

=== Memory Information ===
MemTotal:        3898604 kB
MemFree:         3150632 kB
MemAvailable:    3393312 kB
Buffers:           17772 kB
Cached:           343800 kB

=== Process Information ===
Name:   procfs_read
Umask:  0022
State:  R (running)
Tgid:   2488
Ngid:   0
Pid:    2488
PPid:   287
TracerPid:      0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.9 Building and Booting an Operating System

Pada praktik *Building and Booting an Operating System*, pengguna mempelajari proses penting saat sistem operasi Linux dijalankan. Melalui perintah seperti dmesg | less, kita dapat melihat pesan boot kernel untuk memahami tahapan inisialisasi perangkat keras dan driver. Perintah systemd-analyze digunakan untuk menganalisis waktu boot sistem, sementara systemd-analyze blame dan critical-chain membantu mengidentifikasi proses yang memperlambat boot. File konfigurasi GRUB dapat dilihat menggunakan cat /boot/grub/grub.cfg, sedangkan isi *initramfs* diperiksa dengan lsinitramfs. Pada praktik kernel parameters, pengguna mempelajari bagaimana membaca parameter kernel melalui cat /proc/cmdline serta menambah parameter baru, baik secara sementara di menu GRUB maupun permanen melalui file /etc/default/grub dengan editor nano, lalu memperbaruinya menggunakan sudo update-grub.

2.9.1 Analisis Boot Process

- **Lihat boot messages**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg | less

[9]+  Stopped                 dmesg | less
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

```
[    0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_
DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[    0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=4 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_co
unt=0 WSL_ENABLE_CRASH_DUMP=1
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[    0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[    0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[    0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000f7ffffff] usable
[    0.000000] BIOS-e820: [mem 0x0000000100000000-0x00000001011fffff] usable
[    0.000000] NX (Execute Disable) protection: active
[    0.000000] APIC: Static calls initialized
[    0.000000] DMI not present or invalid.
[    0.000000] Hypervisor detected: Microsoft Hyper-V
[    0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x9a4e24, misc 0xe4bed7b6
[    0.000000] Hyper-V: Nested features: 0x3e0101
[    0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[    0.000000] Hyper-V: Using hypercall for remote TLB flush
[    0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] tsc: Detected 2995.198 MHz processor
[    0.000073] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[    0.000077] e820: remove [mem 0x000a0000-0x000fffff] usable
[    0.000081] last_pfn = 0x101200 max_arch_pfn = 0x400000000
[    0.000103] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[    0.000106] x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
[    0.000143] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[    0.000161] Using GB pages for direct mapping
[    0.000307] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[    0.000310] ACPI: Early table checksum verification disabled
[    0.000315] ACPI: RSDP 0x00000000000E0000 000024 (v02 VRTUAL)
[    0.000320] ACPI: XSDT 0x0000000000100000 000044 (v01 VRTUAL MICROSFT 00000001 MSFT 00000001)
:
```

- **Lihat systemd boot analysis**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ systemd-analyze
Startup finished in 3.483s (userspace)
graphical.target reached after 3.462s in userspace.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ systemd-analyze blame
3.895s motd-news.service
1.435s landscape-client.service
 946ms dev-sdd.device
 829ms snapd.seeded.service
 627ms snapd.service
 530ms systemd-resolved.service
 459ms wsl-pro.service
 350ms rsyslog.service
 299ms dev-hugepages.mount
 295ms dev-mqueue.mount
 289ms systemd-timesyncd.service
 273ms snapd.socket
 273ms sys-kernel-debug.mount
 241ms sys-kernel-tracing.mount
 221ms my_daemon.service
 216ms dbus.service
 201ms systemd-tmpfiles-clean.service
 200ms systemd-journald.service
 200ms systemd-udev-trigger.service
 196ms e2scrub_reap.service
 195ms keyboard-setup.service
 184ms systemd-sysctl.service
 180ms kmod-static-nodes.service
 168ms systemd-journal-flush.service
 159ms modprobe@configfs.service
 159ms systemd-logind.service
 157ms user@1000.service
 146ms modprobe@dm_mod.service
 119ms modprobe@drm.service
 114ms systemd-udevd.service
 113ms modprobe@efi_pstore.service
 108ms systemd-tmpfiles-setup-dev.service
 101ms modprobe@fuse.service
  82ms modprobe@loop.service
  74ms sys-fs-fuse-connections.mount
lines 1-35
[10]+  Stopped                 systemd-analyze blame
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @3.462s
└─multi-user.target @3.462s
  └─snapd.seeded.service @2.063s +829ms
    └─basic.target @1.955s
      └─sockets.target @1.951s
        └─snapd.socket @1.674s +273ms
          └─sysinit.target @1.643s
            └─systemd-resolved.service @1.089s +530ms
              └─systemd-tmpfiles-setup.service @1.012s +48ms
                └─local-fs.target @1.004s
                  └─local-fs-pre.target @1.004s
                    └─systemd-tmpfiles-setup-dev.service @893ms +108ms
                      └─systemd-tmpfiles-setup-dev-early.service @825ms +33ms
                        └─kmod-static-nodes.service @527ms +180ms
                          └─systemd-journald.socket @404ms
                            └─-.mount @352ms
                              └─-.slice @352ms
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat GRUB configuration**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

- **Lihat initramfs**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
/usr/bin/unmkinitramfs: 64: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 57: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 38: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
cpio: premature end of archive
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.10 Operating-System Debugging

Praktik Operating-System Debugging pada Linux bertujuan untuk memahami cara menemukan dan menganalisis kesalahan dalam sistem maupun program. Dengan strace, pengguna dapat melacak system calls yang dijalankan oleh perintah seperti `ls`, serta melihat berapa lama setiap system call berlangsung. Sementara itu, ltrace digunakan untuk memantau panggilan library functions agar programmer mengetahui fungsi-fungsi pustaka apa saja yang dijalankan. GDB (GNU Debugger) berfungsi mendeteksi kesalahan logika atau bug dalam program, seperti pada contoh `buggy.c` yang memiliki kesalahan pada batas array. dmesg digunakan untuk membaca log kernel, memantau error, atau peringatan sistem. Selain itu, Valgrind membantu mendeteksi kebocoran memori, sedangkan perf digunakan untuk menganalisis performa sistem dan memonitor aktivitas CPU. Semua alat ini memungkinkan pengguna memahami perilaku sistem operasi secara mendalam dan memperbaiki masalah dengan efisien.

2.10.1 Using strace

- **Trace system calls**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffdc32cb4e0 /* 27 vars */) = 0
brk(NULL)                               = 0x651a39df5000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x78f81eeb6000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20143, ...}) = 0
mmap(NULL, 20143, PROT_READ, MAP_PRIVATE, 3, 0) = 0x78f81eeb1000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x78f81ee84000
mmap(0x78f81ee8a000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x78f81ee8a000
mmap(0x78f81eea7000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x78f81eea7000
mmap(0x78f81eead000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x78f81eead000
mmap(0x78f81eeaf000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x78f81eeaf000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x78f81ec00000
mmap(0x78f81ec28000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x78f81ec28000
mmap(0x78f81edb0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x78f81edb0000
mmap(0x78f81edff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x78f81edff000
mmap(0x78f81ee05000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x78f81ee05000
```

- **Trace dengan detail**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ strace -v ls
execve("/usr/bin/ls", ["ls"], ["SHELL=/bin/bash", "WSL2_GUI_APPS_ENABLED=1", "WSL_DISTRO_NAME=Ubuntu", "NAME=LAPTOP-GRJL2V1D", "PWD=/tmp/test_di
r", "LOGNAME=danyp", "HOME=/home/danyp", "LANG=C.UTF-8", "WSL_INTEROP=/run/WSL/286_interop", "LS_COLORS=rs=0:di=01;34:ln=01;36"..., "WAYLAND_DIS
PLAY=wayland-0", "LESSCLOSE=/usr/bin/lesspipe %s %"..., "TERM=xterm-256color", "LESSOPEN=| /usr/bin/lesspipe %s", "USER=danyp", "DISPLAY=:0", "S
HLVL=1", "XDG_RUNTIME_DIR=/run/user/1000/", "DEBUGINFOD_URLS=https://debuginf"..., "WSLENV=", "XDG_DATA_DIRS=/usr/local/share:/"..., "PATH=/usr/
local/sbin:/usr/local/"..., "DBUS_SESSION_BUS_ADDRESS=unix:pa"..., "HOSTTYPE=x86_64", "PULSE_SERVER=unix:/mnt/wslg/Puls"..., "OLDPWD=/tmp", "_=/
usr/bin/strace"]) = 0
brk(NULL)                               = 0x6264c84fa000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x77c1ad461000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=2603, st_mode=S_IFREG|0644, st_nlink=1, st_uid=0, st_gid=0, st_blksize=4096, st_blocks=40, st_size=2
0143, st_atime=1760363166 /* 2025-10-13T20:46:06.914040011+0700 */, st_atime_nsec=914040011, st_mtime=1760363166 /* 2025-10-13T20:46:06.90204001
1+0700 */, st_mtime_nsec=902040011, st_ctime=1760363166 /* 2025-10-13T20:46:06.906040011+0700 */, st_ctime_nsec=906040011}) = 0
mmap(NULL, 20143, PROT_READ, MAP_PRIVATE, 3, 0) = 0x77c1ad45c000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=13037, st_mode=S_IFREG|0644, st_nlink=1, st_uid=0, st_gid=0, st_blksize=4096, st_blocks=344, st_size
=174472, st_atime=1760317496 /* 2025-10-13T08:04:56.051264773+0700 */, st_atime_nsec=51264773, st_mtime=1727872852 /* 2024-10-02T19:40:52+0700 *
/, st_mtime_nsec=0, st_ctime=1759843762 /* 2025-10-07T20:29:22.380646443+0700 */, st_ctime_nsec=380646443}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77c1ad42f000
mmap(0x77c1ad435000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x77c1ad435000
mmap(0x77c1ad452000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x77c1ad452000
mmap(0x77c1ad458000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x77c1ad458000
mmap(0x77c1ad45a000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x77c1ad45a000
close(3)                                = 0
```

- **Hitung waktu system calls**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ strace -c ls
Makefile          hello_module.o    main.c  math_ops.c  math_ops.s     output.txt           procfs_read.c    program_static
file_syscalls     info_syscalls     main.i  math_ops.h  my_daemon      process_syscalls     program          simple_shell
file_syscalls.c   info_syscalls.c   main.o  math_ops.i  my_daemon.c    process_syscalls.c   program_dynamic  simple_shell.c
hello_module.c    libmathops.so     main.s  math_ops.o  os_specific.c  procfs_read          program_shared   syscall_test.txt
% time     seconds  usecs/call     calls    errors syscall
------ ----------- ----------- --------- --------- ----------------
 24.63    0.001190          39        30           mmap
 24.01    0.001160          33        35        13 openat
 12.19    0.000589          24        24           close
 10.62    0.000513          22        23           fstat
  7.26    0.000351          35        10           read
  4.22    0.000204          40         5           mprotect
  3.44    0.000166          41         4           write
  2.42    0.000117          58         2           getdents64
  1.61    0.000078          39         2         1 statfs
  1.53    0.000074          37         2         2 access
  1.45    0.000070          23         3           brk
  1.39    0.000067          67         1           futex
  0.97    0.000047          23         2           ioctl
  0.89    0.000043          21         2           pread64
  0.68    0.000033          33         1           munmap
  0.48    0.000023          23         1           set_robust_list
  0.46    0.000022          22         1           prlimit64
  0.46    0.000022          22         1           getrandom
  0.43    0.000021          21         1           arch_prctl
  0.43    0.000021          21         1           set_tid_address
  0.43    0.000021          21         1           rseq
  0.00    0.000000           0         1           execve
------ ----------- ----------- --------- --------- ----------------
100.00    0.004832          31       153        16 total
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Trace dan simpan ke file**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ strace -o trace.txt ls
Makefile          hello_module.o    main.c  math_ops.c  math_ops.s     output.txt           procfs_read.c    program_static    trace.txt
file_syscalls     info_syscalls     main.i  math_ops.h  my_daemon      process_syscalls     program          simple_shell
file_syscalls.c   info_syscalls.c   main.o  math_ops.i  my_daemon.c    process_syscalls.c   program_dynamic  simple_shell.c
hello_module.c    libmathops.so     main.s  math_ops.o  os_specific.c  procfs_read          program_shared   syscall_test.txt
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat trace.txt
execve("/usr/bin/ls", ["ls"], 0x7ffcba8170d0 /* 27 vars */) = 0
brk(NULL)                               = 0x603df3ad8000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x762f35ccd000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20143, ...}) = 0
mmap(NULL, 20143, PROT_READ, MAP_PRIVATE, 3, 0) = 0x762f35cc8000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x762f35c9b000
mmap(0x762f35ca1000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x762f35ca1000
mmap(0x762f35cbe000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x762f35cbe000
mmap(0x762f35cc4000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x762f35cc4000
mmap(0x762f35cc6000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x762f35cc6000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x762f35a00000
mmap(0x762f35a28000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x762f35a28000
mmap(0x762f35bb0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x762f35bb0000
mmap(0x762f35bff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x762f35bff000
mmap(0x762f35c05000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x762f35c05000
close(3)                                = 0
```

2.10.2 Using ltrace

- **Trace library calls**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ltrace ls
strrchr("ls", '/')                                                          = nil
setlocale(LC_ALL, "")                                                       = "C.UTF-8"
bindtextdomain("coreutils", "/usr/share/locale")                            = "/usr/share/locale"
textdomain("coreutils")                                                     = "coreutils"
__cxa_atexit(0x5837f674f880, 0, 0x5837f6767008, 0)                          = 0
getopt_long(1, 0x7fff71a11fa8, "abcdfghiklmnopqrstuvw:xABCDFGHI:"..., 0x5837f6766340, -1) = -1
getenv("LS_BLOCK_SIZE")                                                     = nil
getenv("BLOCK_SIZE")                                                        = nil
getenv("BLOCKSIZE")                                                         = nil
getenv("POSIXLY_CORRECT")                                                   = nil
getenv("BLOCK_SIZE")                                                        = nil
isatty(1)                                                                   = 1
ioctl(1, 21523, 0x7fff71a11dc0)                                             = 0
getenv("TABSIZE")                                                           = nil
getenv("QUOTING_STYLE")                                                     = nil
__errno_location()                                                          = 0x76246c8dd6a0
__errno_location()                                                          = 0x76246c8dd6a0
getenv("TZ")                                                                = nil
reallocarray(0, 100, 208, 0x58381c8446c0)                                   = 0x58381c8446d0
strlen(".")                                                                 = 1
__memcpy_chk(0x58381c849850, 0x5837f675eedc, 2, 2)                          = 0x58381c849850
__errno_location()                                                          = 0x76246c8dd6a0
opendir(".")                                                                = 0x58381c849870
readdir(0x58381c849870)                                                     = 0x58381c8498a0
__errno_location()                                                          = 0x76246c8dd6a0
__ctype_get_mb_cur_max()                                                    = 6
strlen("program_static")                                                    = 14
strlen("program_static")                                                    = 14
```

- **Dengan detail**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ ltrace -c ls
Makefile         hello_module.c  info_syscalls.c  main.i  math_ops.c  math_ops.o  my_daemon.c    process_syscalls    procfs_read.c    program_shared  simple_shell.c
file_syscalls    hello_module.o  libmathops.so    main.o  math_ops.h  math_ops.s  os_specific.c  process_syscalls.c  program          program_static  syscall_test.txt
file_syscalls.c  info_syscalls   main.c           main.s  math_ops.i  my_daemon   output.txt     procfs_read         program_dynamic  simple_shell    trace.txt
% time     seconds  usecs/call     calls      function
------ ----------- ----------- --------- --------------------
 21.28    0.032365         198       163 __errno_location
 16.56    0.025191         198       127 strcoll
 16.17    0.024589         184       133 strlen
 13.48    0.020495         207        99 __ctype_get_mb_cur_max
  8.41    0.012789         155        82 __overflow
  5.71    0.008678        8678         1 setlocale
  4.42    0.006724         186        36 readdir
  3.87    0.005890         173        34 __memcpy_chk
  3.51    0.005335         161        33 fwrite_unlocked
  2.50    0.003806         211        18 memcpy
  0.82    0.001251         156         8 getenv
  0.42    0.000638         159         4 reallocarray
  0.34    0.000517         129         4 __freading
  0.32    0.000491         245         2 fclose
  0.25    0.000377         377         1 opendir
  0.23    0.000343         343         1 closedir
  0.20    0.000301         150         2 fileno
  0.20    0.000298         298         1 ioctl
  0.19    0.000284         142         2 __fpending
  0.18    0.000281         281         1 _setjmp
  0.18    0.000277         138         2 fflush
  0.18    0.000267         267         1 bindtextdomain
  0.17    0.000262         262         1 isatty
  0.11    0.000174         174         1 strrchr
  0.10    0.000158         158         1 getopt_long
  0.10    0.000155         155         1 textdomain
  0.09    0.000143         143         1 __cxa_atexit
------ ----------- ----------- --------- --------------------
100.00    0.152079                   760 total
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

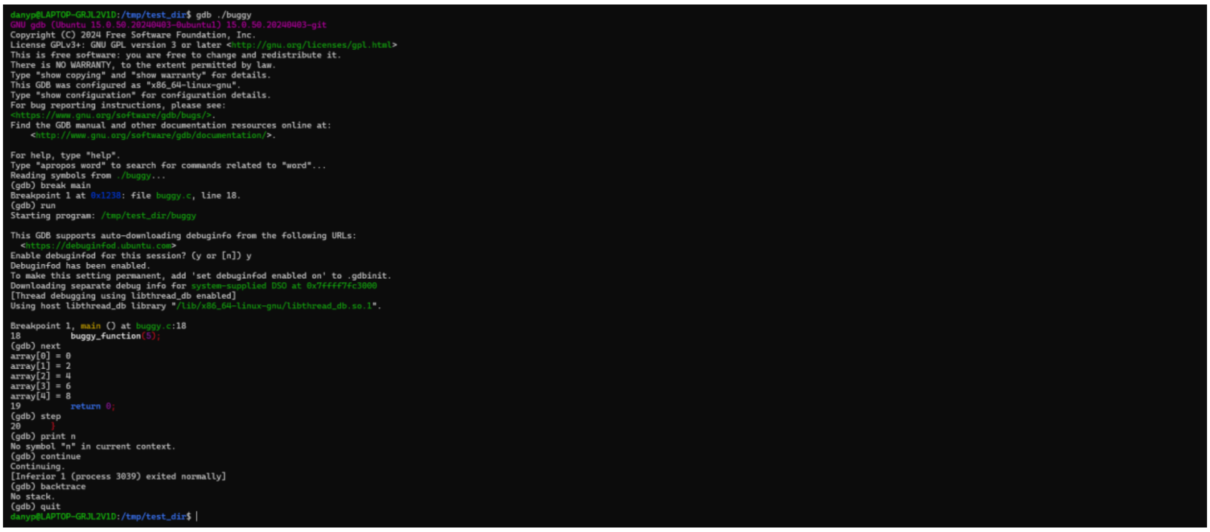
2.10.3 Using GDB

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ nano buggy.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Compile dengan debug symbols**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -g -o buggy buggy.c
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

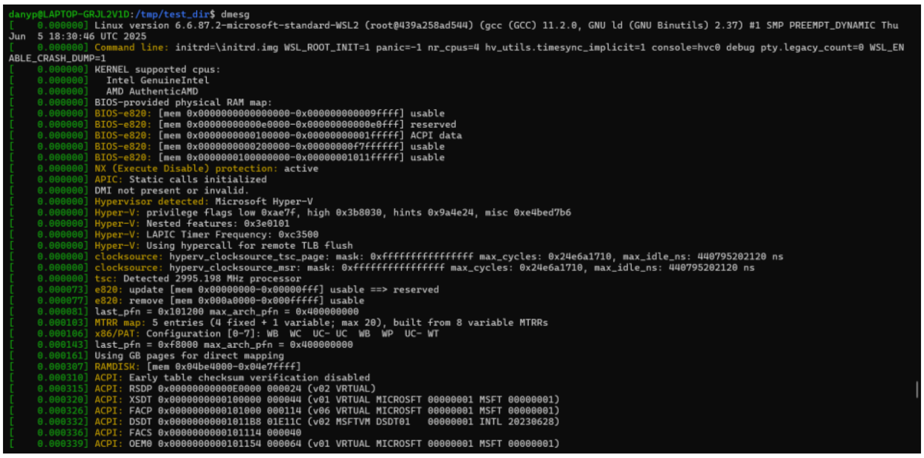
- **Jalankan dengan GDB**



```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 18.
(gdb) run
Starting program: /tmp/test_dir/buggy

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:18
18          buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
19          return 0;
(gdb) step
20      }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 3039) exited normally]
(gdb) backtrace
No stack.
(gdb) quit
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ |
```

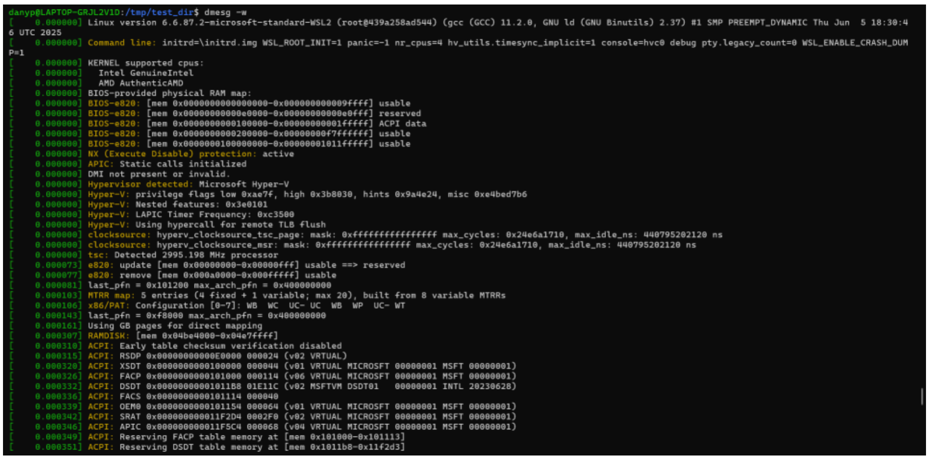
2.10.4 Kernel Debugging dengan dmesg

- **Lihat kernel log**



```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg
[    0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu
Jun  5 18:30:46 UTC 2025
[    0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=4 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_EN
ABLE_CRASH_DUMP=1
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[    0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[    0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[    0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000f7ffffff] usable
[    0.000000] BIOS-e820: [mem 0x0000000100000000-0x00000001011fffff] usable
[    0.000000] NX (Execute Disable) protection: active
[    0.000000] APIC: Static calls initialized
[    0.000000] DMI not present or invalid.
[    0.000000] Hypervisor detected: Microsoft Hyper-V
[    0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x9a4e24, misc 0xe4bed7b6
[    0.000000] Hyper-V: Nested features: 0x3e0101
[    0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[    0.000000] Hyper-V: Using hypercall for remote TLB flush
[    0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] tsc: Detected 2995.198 MHz processor
[    0.000073] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[    0.000077] e820: remove [mem 0x000a0000-0x000fffff] usable
[    0.000081] last_pfn = 0x101200 max_arch_pfn = 0x400000000
[    0.000103] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[    0.000106] x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
[    0.000143] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[    0.000161] Using GB pages for direct mapping
[    0.000307] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[    0.000310] ACPI: Early table checksum verification disabled
[    0.000315] ACPI: RSDP 0x00000000000E0000 000024 (v02 VRTUAL)
[    0.000320] ACPI: XSDT 0x0000000000100000 000044 (v01 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000326] ACPI: FACP 0x0000000000101000 000114 (v06 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000332] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01   00000001 INTL 20230628)
[    0.000336] ACPI: FACS 0x0000000000101114 000040
[    0.000339] ACPI: OEM0 0x0000000000101154 000064 (v01 VRTUAL MICROSFT 00000001 MSFT 00000001)
```

- **Follow kernel log**



```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg -w
[    0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:4
6 UTC 2025
[    0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=4 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUM
P=1
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[    0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[    0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[    0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000f7ffffff] usable
[    0.000000] BIOS-e820: [mem 0x0000000100000000-0x00000001011fffff] usable
[    0.000000] NX (Execute Disable) protection: active
[    0.000000] APIC: Static calls initialized
[    0.000000] DMI not present or invalid.
[    0.000000] Hypervisor detected: Microsoft Hyper-V
[    0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x9a4e24, misc 0xe4bed7b6
[    0.000000] Hyper-V: Nested features: 0x3e0101
[    0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[    0.000000] Hyper-V: Using hypercall for remote TLB flush
[    0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] tsc: Detected 2995.198 MHz processor
[    0.000073] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[    0.000077] e820: remove [mem 0x000a0000-0x000fffff] usable
[    0.000081] last_pfn = 0x101200 max_arch_pfn = 0x400000000
[    0.000103] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[    0.000106] x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
[    0.000143] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[    0.000161] Using GB pages for direct mapping
[    0.000307] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[    0.000310] ACPI: Early table checksum verification disabled
[    0.000315] ACPI: RSDP 0x00000000000E0000 000024 (v02 VRTUAL)
[    0.000320] ACPI: XSDT 0x0000000000100000 000044 (v01 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000326] ACPI: FACP 0x0000000000101000 000114 (v06 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000332] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01   00000001 INTL 20230628)
[    0.000336] ACPI: FACS 0x0000000000101114 000040
[    0.000339] ACPI: OEM0 0x0000000000101154 000064 (v01 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000342] ACPI: SRAT 0x000000000011F2D4 0002F0 (v02 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000346] ACPI: APIC 0x000000000011F5C4 000068 (v04 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000349] ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
[    0.000351] ACPI: Reserving DSDT table memory at [mem 0x1011b8-0x11f2d3]
```

- **Filter by level**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg -l err
[    0.144141] PCI: Fatal: No config space access function found
[    2.142989] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[    2.187696] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[    2.189786] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[    2.191979] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[    2.193889] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[   11.224293] WSL (188) ERROR: CheckConnection: getaddrinfo() failed: -3
[11031.730693] Invalid ELF header magic: != \x7fELF
[11827.851284] Invalid ELF header magic: != \x7fELF
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ dmesg -l warn
[    0.135914] Hyper-V: Disabling IBT because of Hyper-V bug
[    0.176253] PCI: System does not support PCI
[    0.262602] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA
 log.
[    1.994086] pulseaudio[220]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[    2.270177] Failed to connect to bus: No such file or directory
[    2.847600] systemd-journald[43]: File /var/log/journal/1cb2442bdf224e8691d1e1948ea146fb/system.journal corrupted or uncleanly shu
t down, renaming and replacing.
[15130.198315] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.10.5 Memory Leak Detection dengan Valgrind

- **Install valgrind**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo apt install valgrind
[sudo] password for danyp:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
valgrind is already the newest version (1:3.22.0-0ubuntu3).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Buat program dengan memory leak**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ cat > memleak.c << 'EOF'
>  #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
return 0;
}
EOF
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ gcc -g -o memleak memleak.c
```

- **Jalankan dengan valgrind**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ valgrind --leak-check=full ./memleak
==3172== Memcheck, a memory error detector
==3172== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==3172== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==3172== Command: ./memleak
==3172==
==3172==
==3172== HEAP SUMMARY:
==3172==     in use at exit: 400 bytes in 1 blocks
==3172==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==3172==
==3172== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==3172==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==3172==    by 0x10915E: main (memleak.c:3)
==3172==
==3172== LEAK SUMMARY:
==3172==    definitely lost: 400 bytes in 1 blocks
==3172==    indirectly lost: 0 bytes in 0 blocks
==3172==      possibly lost: 0 bytes in 0 blocks
==3172==    still reachable: 0 bytes in 0 blocks
==3172==         suppressed: 0 bytes in 0 blocks
==3172==
==3172== For lists of detected and suppressed errors, rerun with: -s
==3172== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

2.10.6 Performance Analysis

- **CPU profiling dengan perf**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
linux-tools-common is already the newest version (6.8.0-85.85).
linux-tools-generic is already the newest version (6.8.0-85.85).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Monitor system calls**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

  You may need to install the following packages for this specific kernel:
    linux-tools-6.6.87.2-microsoft-standard-WSL2
    linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

  You may also want to install one of the following packages to keep up to date:
    linux-tools-standard-WSL2
    linux-cloud-tools-standard-WSL2
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

- **Top-like interface**

```
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

  You may need to install the following packages for this specific kernel:
    linux-tools-6.6.87.2-microsoft-standard-WSL2
    linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

  You may also want to install one of the following packages to keep up to date:
    linux-tools-standard-WSL2
    linux-cloud-tools-standard-WSL2
danyp@LAPTOP-GRJL2V1D:/tmp/test_dir$
```

BAB III

PENUTUP

3.1 KESIMPULAN

Berdasarkan seluruh rangkaian praktikum Sistem Operasi yang telah dilakukan, dapat disimpulkan bahwa Linux memberikan pemahaman menyeluruh mengenai cara kerja inti dari sistem operasi, mulai dari layanan dasar hingga manajemen kernel. Melalui berbagai praktik, mahasiswa mempelajari fungsi utama sistem operasi seperti pengelolaan proses, sistem berkas, layanan sistem, serta pemanggilan sistem (system calls) yang menjadi jembatan antara aplikasi dan kernel. Selain itu, praktik tentang systemd, linker & loader, serta struktur OS memperlihatkan bagaimana Linux bekerja secara modular dan berlapis untuk menjaga efisiensi serta stabilitas sistem. Mahasiswa juga memahami bagaimana program dapat bersifat spesifik terhadap sistem operasi tertentu karena perbedaan format biner dan API. Praktik kernel module, proses booting, hingga debugging menggunakan alat seperti strace, ltrace, gdb, dan valgrind melatih kemampuan analisis serta troubleshooting. Secara keseluruhan, praktikum ini memperkuat pemahaman konseptual dan praktis tentang arsitektur, manajemen sumber daya, serta mekanisme internal sistem operasi Linux yang menjadi fondasi penting dalam dunia komputasi dan pengembangan perangkat lunak.